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Evaluating vehicle fleet electrification against net-zero targets in scooter-dominated road transport

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ABSTRACT

Vehicle electrification is recognized as a pathway toward net-zero emissions. However, it is unclear if the existing timeline for banning internal combustion engines is sufficient to meet the climate goal, especially for the regions where two-wheelers dominate road transport. We develop a novel bootstrap-based modeling framework to examine fleet turnover and future greenhouse gas emissions during the decarbonization transformations. With the highest scooter density globally, Taiwan is selected as a case study. The model thoroughly captures the key road characteristics, including an inverse U-shaped relationship between scooter ownership and income. Under Taiwan's 2050 Net-Zero Pathway, transitioning to electric vehicles (EVs) will bring significant climate benefits but still fall short of the target. While the growing EV adoption shifts the carbon burden from transport to the power generation sector, these increased upstream emissions will eventually be counterbalanced by clean grid development following the power sector roadmap by 2050.

1. Introduction

The 2015 Paris Agreement sets out a framework for continuous climate action that aims to limit global warming to below 2 °C above preindustrial levels while pursuing efforts to limit it to 1.5 °C (United Nations Framework Convention on Climate Change (UNFCCC), 2020). To achieve this long-term temperature goal, all countries—especially the biggest emitters—must commit to their national climate pledges. For example, the United States should commit to reduce emissions by 50 % to 55 % from 2005 levels by 2030 and to meet net-zero targets by 2050; China should pledge to peak CO_2 emissions by 2026 and to achieve carbon neutrality by 2060; the European Union, by 2030, should reduce its net greenhouse gas (GHG) emissions by at least 55 % compared to 1990 level according to European Climate Law (Keohane, 2020). The global consensus on "Net-Zero Emissions" is considered ambitious but will greatly help prevent climate disasters if fully realized. The transportation sector contributes about 27 % of global GHG emissions (International Energy Agency, 2019), implying that the decarbonization of the transportation sector—especially road transport—remains vital to a net-zero global economy. Strategies to decarbonize the transportation sector include technology efficiency improvements, low-carbon alternative fuels, and vehicle electrification. Among all of the possible approaches, clean electrification is recognized to be the single most crucial pathway (International Council on Clean Transportation, 2020). Thus, a growing coalition of countries and cities is committing to phasing out the sales of new internal-combustion engine vehicles (ICEVs) and putting more effort into promoting

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electric vehicle (EV) deployment.

Considering that fossil fuels are still the dominant energy source nowadays, the benefits of EVs for addressing climate change remain unclear. Life-cycle assessment (LCA) is a widely-applied method to compare the emissions per distance driven among different vehicle technologies (Wang et al., 2018). Nordelöf (Nordelöf et al., 2014) and Marmiroli (Marmiroli et al., 2018) investigated 79 and 44 articles regarding LCA studies of EVs, respectively. The majority of the reviewed studies shared the same conclusion that the carbon intensity of electricity generation is the key factor affecting the environmental benefits of vehicle electrification. To assess the aggregate energy demand and GHG emissions from road transport, several studies incorporated bottom-up vehicle fleet models with plentiful four-wheel details such as historical vehicle stock growth, vehicle distance traveled, fuel efficiency improvement, and advanced vehicle technology adoption. Vehicle fleet research and modeling originated in the U.S. market; the highly-cited researches include the MIT FLEET model (Bandivadekar, 2008), the LEVERS model (Yang et al., 2009), and the GREET model (Wang, 1999). These models have been broadly adopted to forecast vehicle fleet energy and emissions to understand the transport policy impacts in different countries. China, as the world's largest automobile market and the largest carbon emitter, vehicle fleet modeling and emissions reduction strategies have received increasing research attention. Economic elasticity methods were shown to have good short-term prediction performance in China's vehicle stocks and the associated emissions (He et al., 2005; Wang et al., 2007). For long-term vehicle ownership forecasting, the Gompertz function (S-curve) is frequently used to depict how private vehicle stock grows as the economy (such as GDP per capita, income per capita, and vehicle purchasing power) develops (Hsieh et al., 2018; Huo et al., 2007; Huo and Wang, 2012; Wu et al., 2014). Other economic characteristics like urban population density are also responsible for the on-road non-private vehicle dynamics (Hao et al., 2011; Peng et al., 2018).

When assessing the road transportation systems, most of the above-mentioned studies excluded two-wheel vehicles, which may introduce significant bias in the calculations—especially for South-East Asia countries where scooters (or motorcycles) are the favored road transport mode. Compared to four-wheel vehicles, vehicle fleet model development research on two-wheelers is still relatively lacking. Gompertz model (S-shaped curve) is usually used to project vehicle ownership; however, it gave the unreasonable curve fitting for the motorcycle ownership modeling in East Asia (Hsu, 2005). More recently, several studies pointed out that motorcycle ownership would grow during the beginning stages as the income increases, but instead of approaching a saturation level, it would start declining once a threshold level is exceeded—the so-called "Motorcycle Kuznets Curve," an inverse-U relationship (Chu et al., 2022; Nishitateno and Burke, 2014). To the best of our knowledge, modeling the future trends of on-road vehicles – including two-wheelers – and the corresponding life-cycle carbon emissions in response to a net-zero roadmap have not yet been conducted. Considering the global transportation system is undergoing a far-reaching transition toward net-zero emissions, it is of timely importance to have such a model that comprehensively characterizes the key features of road transport.

With the world's highest density of two-wheeler ownership (Ministry of Transportation and Communications, 2021), Taiwan is selected as a case study when we develop such modeling capabilities for examining long-term vehicle fleet trends and GHG emissions under various policies. Although not a signatory to the Paris Agreement, Taiwan has also set ambitious reduction targets for its GHG emissions. The Taiwanese government has demonstrated its determination to limit global warming since the Greenhouse Gas Reduction and Management Act (the GHG Act) took effect in 2015 (INDC, Executive Yuan, 2015). In response to the rising global ambitions on climate change mitigation, Taiwan has proactively amended the GHG Act to the Climate Change Response Act—with the inclusion of the 2050 net-zero emission target (Environmental Protection Administration, 2022). The ongoing net-zero transition will highlight a rapid increase in renewables; however, other sectors such as transportation will also be deeply involved. In Taiwan, the transportation sector contributes 16 % of total energy use, 13 % of fossil carbon emissions, 32 % of particulate matter ($PM_{2.5}$) emissions, and 51 % of nitrogen oxides (NO_X) pollution (Bureau of Energy, 2019; Environmental Protection Administration, 2019). Considering road transport has the great potential to make significant contributions to sustainability, the recently published "Taiwan's Pathway to Net-Zero Emissions in 2050 (the 2050 Net-Zero Pathway)" clearly sets the timeline for vehicle electrification. Firstly, all the operating buses should be electric by 2030; secondly, all the new sales of private cars and scooters in 2040 should be electric (National Development Council, 2022). However, in Taiwan, where 98 % of the energy consumption relies on imports and 80 % of the electricity generation comes from fossil fuels, vehicle electrification offers clear energy security benefits but unclear climate benefits.

Despite decades of development in vehicle fleet modeling and LCA, only a limited number of related studies have examined road transport in Taiwan. By assuming different growth patterns of the on-road vehicle stocks, Chung and Chang (Chung and Chang, 2016) indicated that the climate change mitigation goal of 50 % carbon reductions by 2050 could be achieved under zero or even negative vehicle stock growth rates and with no ICEVs on the road—focusing only on the direct emissions. Chung-Hua Institution for Economic Research (CIER) conducted a localized LCA and found that the well-to-wheel (WTW) per-km energy consumption and carbon emissions per distance driven of EVs are significantly lower than ICEVs by 57 % and 53 %, respectively (CIER et al., 2016). While the major novelty of this study is in the on-road vehicle fleet model development, the uncertainties in vehicle adoption trajectory are also quantified to improve the existing research methodologies—using single-point estimates for projections, which could lead to a "flaw of averages" and obscure future uncertainty. Moreover, fuel life-cycle (i.e., WTW) energy demand and carbon emissions of Taiwan's future road transport are also analyzed, for the first time, under the 2050 Net-Zero Pathway. Three scenarios are designed to examine the GHG emissions mitigation potential exclusively by vehicle electrification as well as the deployment synergy between EV and renewable energy: reference (*REF*) scenario, EV deployment (*EV*) scenario, and EVs with cleaner power grids (*EV-REN*) scenario.

2. Methods and data

2.1. Model framework and overview

Fig. 1 illustrates the overview of the Taiwan Vehicle Fleet Model. The model generates three main sequential outputs – Vehicle stock, Energy consumption, and GHG emissions. We classify road vehicles in Taiwan into eight vehicle types: private cars, scooters, business cars, taxis, heavy-duty trucks, light-duty trucks, urban buses, and non-urban buses, owing to their different stock growth rates, use intensity, and potential to be electrified. Three vehicle technology and fuel combinations focused on in the study are ICEV-Gasoline, ICEV-Diesel, and EV—representing 99 % of Taiwan's vehicles operating on roads (Ministry of Transportation and Communications, 2021). Note that only battery electric vehicles (BEVs) are modeled as EVs in the present work, while plug-in hybrid electric vehicles (PHEVs) and fuel cell electric vehicles (FCEVs) are excluded. This is because the 2050 Net-Zero Pathway sets clear targets only for BEV deployment, and it is hard to project the future trajectories of PHEV and FCEV adoption.

Vehicle stocks are driven by socioeconomic factors (e.g., income, Gini index, population density, population, gross domestic product (GDP)) and correspond to the volume portion of the vehicle impacts identity. Uncertainty in the parameterization of vehicle ownership modules makes forecasting these trends challenging. Instead of using single-point estimates, we apply a statistical technique – the bootstrap method (see Section 2.6) – to build up the sampling distribution of module parameters, characterize this inherent uncertainty, and establish the confidence intervals for vehicle stock forecasting. The direct energy consumption from the vehicle operation stage (i.e., pump-to-wheels stage; PTW) is calculated as the product of the vehicle stock, vehicle distance traveled, and fuel consumption rate. While vehicle distance traveled input corresponds to the use portion of the vehicle impacts identity, fuel consumption rate and powertrain mix together correspond to the efficiency portion. The energy demand and GHG intensities of the upstream fuel production (i.e., well-to-pump stage; WTP) are mostly derived from GREET-Taiwan, which is developed based on Argonne's U.S. GREET (Greenhouse gases, Regulated Emissions, and Energy use in Transportation) model to reflect specific fuel production pathways in Taiwan. The overall Taiwan electric grid carbon intensity is calculated as the ratio of GHG emissions from electricity generation and gross electricity generation, which will decrease as renewable energy continues to grow.

2.2. *Private car and* scooter *modules*

2.2.1. Stock and sales

With the government survey of Taiwanese families with different income levels owning cars and scooters (Directorate General of Budget, Accounting and Statistics, 2020), it is confirmed that people with higher disposable income are more likely to own cars but not always likely to own scooters. As income increases, the propensity to own a car follows an S-shaped curve, while the probability of owning a scooter reveals an inverted U-shaped pattern. The private car ($\widehat{PC}$) and scooter ($\widehat{PS}$) stocks are calculated by integrating the product of the disposable income distribution (I) and the probability that an individual with that income level owns a private car or scooter (g or γ) across the entire range of disposable incomes (x), as shown in Eqs. (1) and (2).

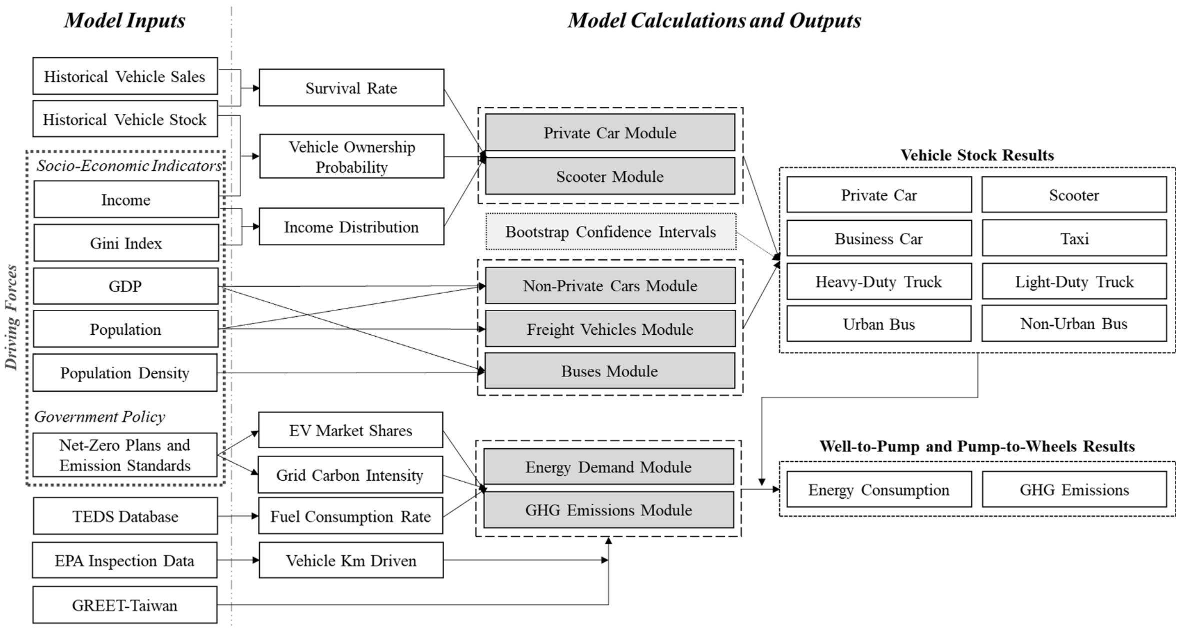


Fig. 1. Methodology framework of the Taiwan Vehicle Fleet Model.

$$\widehat{PC}_i = P_i \int_0^{\infty} I_i(x) g(x) d(x) \tag{1}$$

$$\widehat{PS}_i = P_i \int_0^{\infty} I_i(x) \gamma(x) d(x) \tag{2}$$

where.

$\widehat{PC}_i, \widehat{PS}_i$ = Total private car stock or scooters in year i.

P_i = Population in year i; the future population projection is taken from the middle estimates provided by National Development Council (2020).

x = Disposable income of the incremental individual at constant 2016 New Taiwan dollar (NTD).

I_i = Disposable income distribution function for year i (Section 2.2.2).

g, γ = Relationship between the probability of owning a private car or scooter and the per-capita disposable income (Section 2.2.3).

Private car or scooter sales (S) are calculated by combining new-growth and replacement purchases. New-growth purchases (S^N) represent the spread of vehicle ownership due to rising income, while replacement purchases (S^R) are associated with the sales for scrapped vehicles.

$$S_i = S_i^N + S_i^R \tag{3}$$

$$S_i^N = \widehat{PV}_i - \widehat{PV}_{i-1} \tag{4}$$

$$S_i^R = \sum_{j=1}^{i} S_{i-j} \left(SR_{i,j-1} - SR_{i,j} \right) \tag{5}$$

where.

$\widehat{PV}_i$ = Private vehicle stock in year i;.$\widehat{PV}_i = \{\widehat{PC}_i, \widehat{PS}_i\}$

$SR_{i,j}$ = Survival rate in year i as a function of vehicle age j.

Survival rate describes the relationship between the survival ratio of vehicles (i.e., the share of vehicles remaining in use) and vehicle age (Bento et al., 2018; Zheng et al., 2019). Due to a lack of first-hand scrappage data in Taiwan, this study derives survival patterns based on vehicle registration and population breakdown by age (Ministry of Transportation and Communications, 2021). We use a two-parameter logistic function (Eq.(6)) to simulate the survival rates of private cars and scooters, as plotted in Fig. 2(a).

$$SR_{i,j} = 1/(1 + a\exp(b \times j)) \tag{6}$$

2.2.2. Disposable income distribution (I)

Various probability density functions have been proposed to describe income distribution and inequality. The most commonly used log-normal, gamma, Weibull, log-logistic (Fisk), and Beta distributions are selected and fit to the income data sets from 2008 to 2019 in Taiwan (Directorate General of Budget, Accounting and Statistics, 2020). We calculate the mean squared error (MSE) as a goodness-of-fit measure for each distribution to evaluate the relative performance; the results show that the log-logistic function is the best-fitting model. Thus, we approximate a continuous income distribution of Taiwan with Eq. (7).

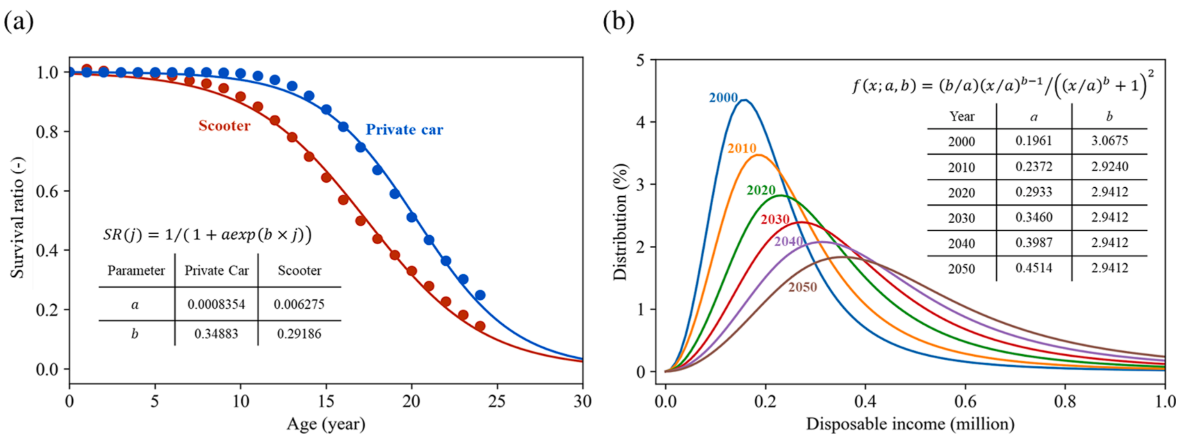


Fig. 2. Key functions in Taiwan's private car and scooter modules: (a) Survival rates; (b) Income distribution with log-logistic distribution assumption, 2000–2050.

$$f(x;a,b) = (b/a)(x/a)^{b-1} \Big/ \left((x/a)^b + 1\right)^2 \tag{7}$$

where.

x = Disposable income per capita.

a, b = Scale and shape parameters of the distribution.

Given the historical and projected mean disposable income per capita (M) and Gini coefficient (G) summarized in Table 1, we solve Eqs. (8) and (9) analytically and simulate the income distribution in a given year (Fig. 2(b)). Due to the lack of future trend studies on the income distribution in Taiwan, linear regression is adopted to describe the historical data from 2010 to 2020 (Directorate General of Budget, Accounting and Statistics, 2020) and extrapolate to the future through 2050.

$$G = 1/b \tag{8}$$

$$M = a\pi/(bsin(\pi/b)) \tag{9}$$

2.2.3. *Ownership probability function (g, γ)*

Income is the main driving force for owning a private vehicle (Dargay and Gately, 1999). For private car ownership, we employ the Gompertz function (Eq. (10)) to estimate its S-shaped relationship with disposable income, like most other motorizing countries. For scooter ownership, instead of applying negative binomial panel regression (Chu et al., 2022), we utilize the Gamma distribution (plus a constant term) (Eq. (11)) to describe the inverse U-shaped relationship between the probability of owning a scooter and income. Gamma distribution is a continuous analog of the negative binomial distribution, relieving the integer constraint on input variables and thus more suitable for the ownership modeling in this study. A constant variable is added to the Gamma distribution to reflect that scooters have been widely considered a necessity in Taiwan (Huang et al., 2009; Institute of Transportation and Chang, 2001). Mathematically, we find that the Gamma distribution plus a constant term has a better fitting performance than the distribution itself alone. Fig. 3 depicts the bootstrapping results of fitting Gompertz and Gamma distributions to the ownership probability data.

$$g(x) = \gamma exp(\alpha exp(\beta x)) \tag{10}$$

$$\gamma(x) = \beta^{\alpha} x^{\alpha-1} exp(-\beta x)/\Gamma(\alpha) + C \tag{11}$$

where.

x = Disposable income per capita.

α, β = Shape and scale parameters of the distribution.

γ = Eventual probability of owning a private car at very high-income levels.

C = Constant term to better describe the scooter ownership.

The private car and scooter ownership probability functions are expected to vary across counties depending on many factors, including population density, car/scooter dependency, mobility levels across different transport modes, road infrastructure development, and consumer behavior. Fig. 4 shows the historical relationship between ownership level versus per-capita disposable income at a county level. We adopt the method proposed by Hsieh et al. (Hsieh et al., 2022) and disaggregate the national-level forecasts of vehicle stocks by counties based on each county's distinct ownership and income relationship.

2.3. ***Non-**private **cars module***

The non-private cars include taxis and business cars owned by companies. Taxis are highly regulated in Taiwan, with regulators determining the ceiling on the number of taxi licenses granted for different counties (Article 91-2 of Regulations for Automobile Transportation Operators, 2021). Consequently, the taxi number has remained relatively stable over the past ten years—ranging from 87 to 92 thousand (Ministry of Transportation and Communications, 2021). On the other hand, the business car stock in Taiwan has developed as an S-shaped curve, gradually approaching the saturation level in recent years. Thus, we use the Gompertz function (Eq.

Table 1
Historical and projected per-capita disposable income, per-capita GDP, Gini index, and total population in Taiwan.

Year	Disposable income per capita (million)	GDP per capita (million)	Gini coefficient (-)	Population (million)
2005	0.262	0.530	0.340	22.77
2010	0.274	0.608	0.342	23.16
2015	0.311	0.727	0.338	23.49
2020	0.370	0.839	0.340	23.56
2025	0.403	0.953	0.276	23.44
2030	0.447	1.072	0.271	23.20
2035	0.492	1.191	0.266	22.79
2040	0.536	1.310	0.261	22.18
2045	0.581	1.429	0.256	21.35
2050	0.626	1.548	0.251	20.37

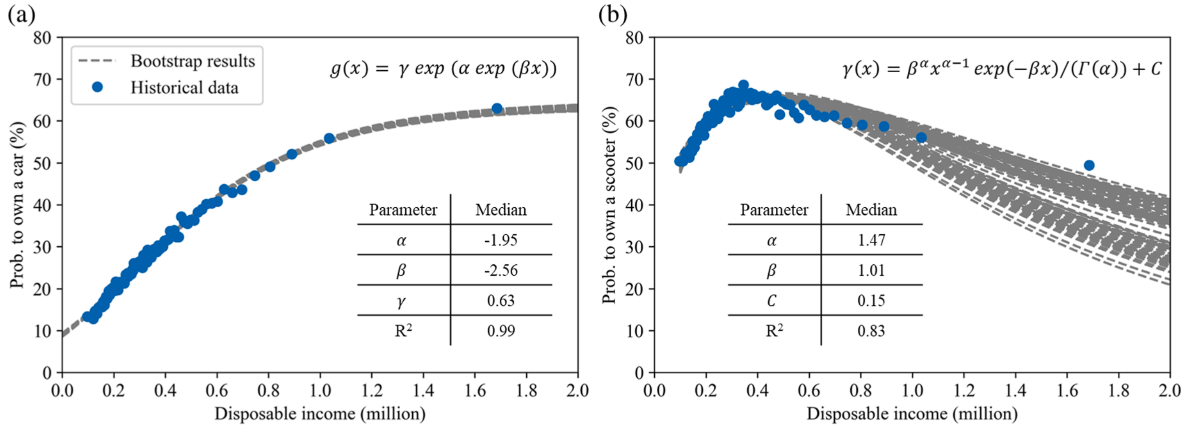


Fig. 3. Ownership probability function fitting with bootstrapping to disposable income per capita (constant 2016 NTD) in Taiwan: (a) Gompertz function $g(x)$ for private cars; (b) Gamma distribution function (plus a constant term)$\gamma(x)$ for scooters.

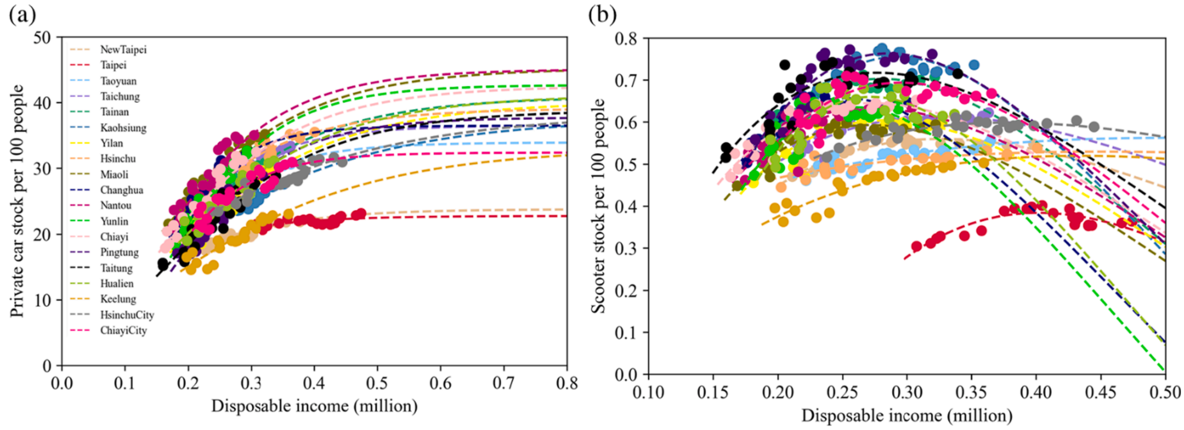


Fig. 4. Private car and scooter ownership levels versus per capita disposable income at the county level from 1998 to 2020.

(9)) to describe the relationship between business car stock and per-capita GDP, with the parameter uncertainty identified using the bootstrap method (Fig. S5(a)). The total number of business cars is estimated based on the projected GDP per capita given in Table 1. The annual sales are calculated using Eq. (3)-(6) with the survival pattern obtained from the annual data of vehicle registration and population breakdown by age (Ministry of Transportation and Communications, 2021). Non-private cars are found to be scrapped at an age much lower than private cars and scooters in Taiwan.

2.4. Freight vehicles module

Diesel prices, GDP, GDP per capita, the natural logarithm of GDP per capita, population, and national area were identified as strong drivers of the road freight service demand (Mulholland et al., 2018). To avoid overfitting, we use the Akaike criterion information (AIC) to compare different linear models for all possible combinations of predictor variables and determine which one best fits the data (i.e., lowest AIC scores) (Portet, 2020). We find that the model with population and the natural logarithm of GDP per capita parameters has the lowest AIC values; Fig. 5(b) shows the linear regression with bootstrapping between the total number of trucks in Taiwan and the natural logarithm of national GDP per capita. The future freight vehicle stock is forecasted based on the projected population and per-capita GDP in Table 1.

Truck stocks are further classified into heavy-duty and light-duty trucks, which are estimated as a product of the total number of trucks and the corresponding stock shares. Over the past decade, the share of heavy trucks in the total truck stock has experienced linear growth (Ministry of Transportation and Communications, 2021). Due to the lack of data, we assume these linear trends will continue toward 2050.

2.5. Buses **module**

Buses are mass transit vehicles with the essential feature of carrying many people in the same vehicle. Bus ownership was assumed

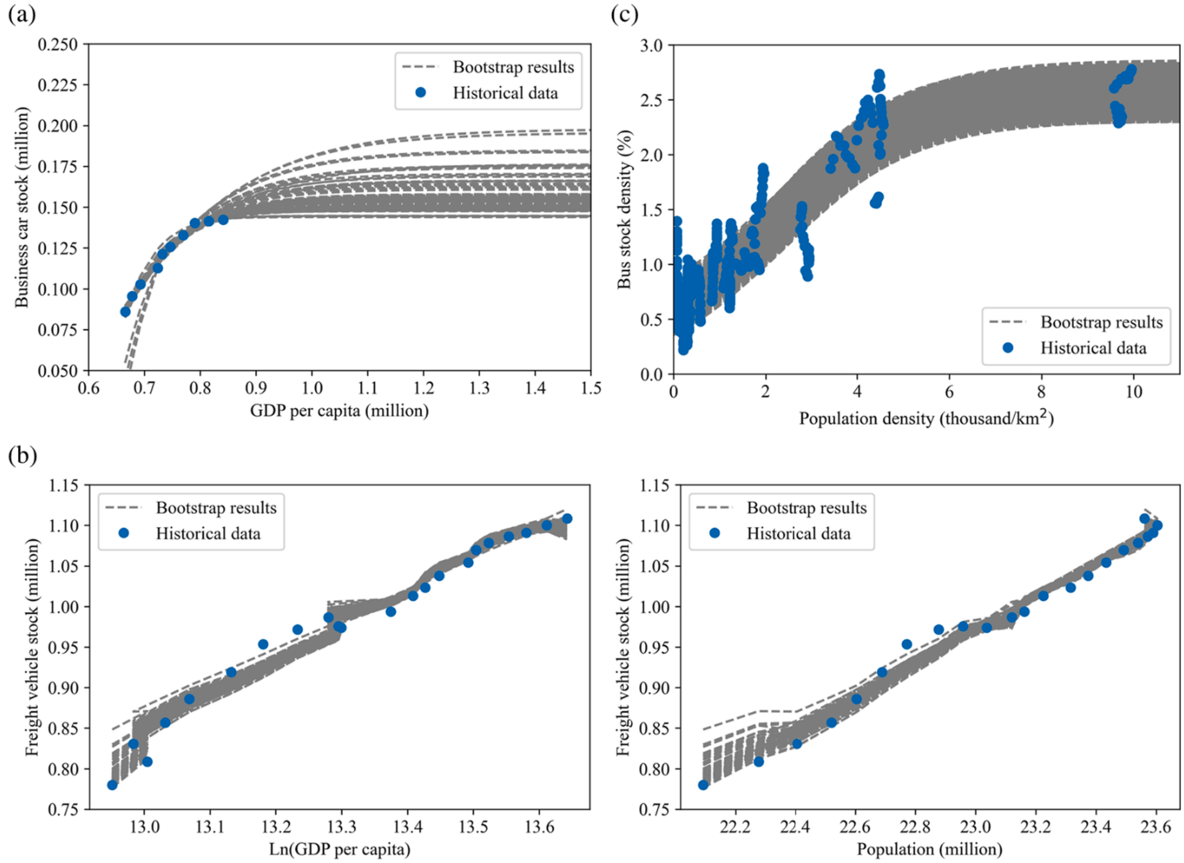


Fig. 5. Vehicle stock function fitting with bootstrapping: (a) Business cars; (b) Freight vehicles; (c) Buses.

linearly related to population and eventually reached saturation levels in the national capital (Peng et al., 2018). However, we find that bus ownership at the county level in Taiwan correlates more with the population density than the population. As population density increases, the county-level bus stocks per thousand people show an S-shaped pattern. We use the logistic distribution best fitting the data among the various sigmoid functions to estimate bus ownership in Taiwan. Fig. 5(c) shows the fitting curves with bootstrapping to the bus ownership and population density data at the country level in Taiwan from 1998 to 2020.

Considering the fact that the government has already set the full electrification target for urban buses (but not for the other types of buses, such as tour buses) (National Development Council, 2022), we further classify bus stocks into urban and non-urban buses. The stock share of the urban bus in Taiwan has generally remained at 49 % in the last decade. Therefore, based on the projected bus ownership, population, and fixed stock share, we estimate the bus stocks by type.

2.6. Bootstrap confidence intervals

Bootstrap is a widely-used computational approach to derive estimates of standard errors and confidence intervals for any statistics or model parameters (Kulesa et al., 2015). Bootstrapping involves repeatedly drawing independent samples from observed data with replacements to create bootstrap samples. In this study, we take 100 bootstrap samples from the datasets used in each vehicle type module; since we draw repeated data with replacement, the same data point may appear multiple times in a single bootstrap sample. By performing regression analysis on each bootstrap sample separately, we obtain 100 bootstrapped values for a model parameter—which we use to estimate the confidence intervals (i.e., sampling uncertainty) around regression coefficients. Note that the future is full of uncertainty and not predictable. Bootstrapping is applied to characterize the uncertainty associated with estimated parameters – rather than an unexpected disruption – to better understand the future.

2.7. Energy demand and GHG emissions modules

The well-to-wheels (WTW) analysis is subdivided into the well-to-pump (WTP) and the pump-to-wheel (PTW) stages. WTP refers to the periods starting from fuel/energy feedstock production to fuel/energy production, i.e., from the extraction of primary energy sources to the production, transmission, and distribution (to gas stations or charging piles) of the energies. PTW refers to the vehicle operation stage where fuel/energy is consumed to power vehicle wheels. Direct energy use (E_{PTW}) of the on-road vehicle fleet in year i

is calculated by the vehicle stock (*V*), fuel consumption rate (*FCR*), and vehicle kilometer traveled (*VKT*) using Eq. (12).

$$E_{PTW,i} = \sum_{k} E_{PTW,i,k} = \sum_{k}\sum_{j} V_{i,j}\cdot S_{i,j,k}\cdot FCR_{i,j,k}\cdot VKT_{i,j} \tag{12}$$

where.

$V_{i,j}$ = Stock of vehicle type *j* in year *i*.

$S_{i,j,k}$ = Stock share of vehicle type *j* powered by fuel/energy *k* in year *i*. For ICEVs (*j* = cars and light-duty trucks), the share of gasoline vehicles in the total vehicle stock has experienced a continuous decline. Due to the lack of data, we assume these trends will continue linearly toward 2050. For EVs (*j* = cars, scooters, and buses), the market shares are explored in the scenario analysis (Section 2.8).

$FCR_{i,j,k}$ = Fuel consumption rate of vehicle type j powered by fuel/energy k in year i. The county-level data for ICEVs are obtained from Taiwan Emission Data System (TEDS) 11.0[2] (Environmental Protection Administration, 2019), with the improvement rates provided by the previous studies (Peng et al., 2018; Wang et al., 2017). As there are no reliable data for EVs, the national sales-weighted average FCRs in 2020 are used and assumed to stay constant toward 2050 (Bureau of Energy, 2022). Note that the anticipated FCR improvements in EVs are usually negligible compared to the other propulsion systems in the literature (Argonne National Laboratory, 2022; Heywood et al., 2015). The historical and projected FCRs are summarized in Table 2.

$VKT_{i,j}$ = Vehicle kilometers traveled per vehicle by vehicle type *j* in year *i*. The county-level VKT data of scooters are retrieved from the regular exhaust inspection records provided by the EPA (Environmental Protection Administration, 2021). For all the other vehicle types, the VKT levels are taken from TEDS 110.0 (Environmental Protection Administration, 2019). The details of VKT levels are shown in Table 2. Due to the lack of future trend studies on vehicle use intensity, this study assumes constant VKT throughout 2050. The effects of this assumption on the carbon emission outcomes are detailed in Section 3.4.

The WTW energy demand (E_{WTW}) of the vehicle fleet in year *i* is then determined by using Eq. (13). And the corresponding WTW GHG emissions of vehicles in year *i* are calculated as the product of the energy demand and GHG emissions intensities, as shown in Eq. (14).

$$E_{WTW,i} = \sum_{k} E_{WTW,i,k} = \sum_{k}\left(E_{PTW,i,k}\bullet EI_{WTP,i,k} + E_{PTW,i,k}\right) \tag{13}$$

$$GHG_{WTW,i} = \sum_{k} GHG_{WTW,i,k} = \sum_{k}\left(E_{PTW,i,k}\bullet EF_{WTW,i,k}\right) \tag{14}$$

where.

$EI_{WTP,i,k}$ = Energy intensities of fuel/energy *k* in year *i* during the fuel production and feedstock production stages.

$EF_{WTW,i,k}$ = WTW GHG emissions factors of fuel/energy *k* in year *i*.

The energy intensities and GHG emissions factors of fuels and energies mostly come from the simulation results calculated by the GREET-Taiwan model, except for the grid carbon intensity derived from the data on the electric power plants (Environmental Protection Administration, 2020; Taiwan Power Company, 2020). The development details for the GREET-Taiwan model were previously documented by the Institute of Transportation in Taiwan (Institute of Transportation, 2013). In this study, we update the key parameters (such as the share of crude oil sources) and expand the GREET-Taiwan model to include all the vehicle types on the road. Regarding the electric grid emission factors (i.e., fuel production stage), we follow the European Commission's definition (European Environment Agency, 2021) and calculate it as the ratio of carbon emissions from electricity generation and gross electricity production. We estimate that the WTW GHG emissions of gasoline, diesel, and electricity are 2.85 kgCO2eq/L (PTW 75 %), 3.30 kgCO2eq/L (PTW 82 %), and 0.552 $kgCO_2eq/kWh$ (PTW 0 %), respectively. Fig. 6 summarizes the fleet-average WTW GHG emission results per distance driven by different vehicle technologies in 2020, calculated by the updated GREET-Taiwan model and the carbon intensity of electricity generation assessed in this study.

2.8. Scenario ***design for 2050 Net-Zero Pathway***

The EV market penetration projection is uncertain and highly affected by evolving government policies. Our base scenario, "EV deployment scenario (i.e., *EV* scenario)", assumes full implementation of Taiwan's current vehicle electrification policy. Under the 2050 Net-Zero Pathway, the roadmap for the road transport sector sets an aggressive timeline to electrify all the operating urban buses by 2030 and all the new sales of cars (both private and non-private) and scooters in 2040. The projected EV growth trajectories for cars, scooters, and urban buses are shown in Fig. 7—demonstrating the government's up-to-date commitment to electrification. To assess the GHG emissions impacts of the vehicle electrification plan, we compare the *EV* scenario with an alternative scenario called the "reference scenario (i.e., *REF* scenario)." The *REF* scenario represents a future in which all the existing EV promotion policies are dropped, and nearly all of the on-road vehicles are ICEVs powered by gasoline or diesel. In the *REF* scenario, the emission changes are mainly driven by the vehicle stock market and ICEV fuel efficiency developments.

In both the *EV* and *REF* scenarios, the carbon emission intensity of the electricity generation is assumed to stay the same after the base year 2020. However, the grid emission intensity is expected to decrease from now to 2050 to achieve a net-zero future. To study

[2] TEDS is reviewed every-three years and updated by the Taiwan Environmental Protection Agency (EPA).

Table 2
Annual vehicle kilometer traveled (VKT; km/yr) and fuel consumption rates (L/100 km or kWh/100 km) for different types of vehicles at the national level in Taiwan.

Vehicle Type	Annual VKT	Fuel Type	2015	2020	2025	2030	2035	2040	2045	2050
Private Car	12,179	Gasoline	10.7	9.7	9.0	8.3	8.0	7.7	7.6	7.5
	14,967	Diesel	8.7	7.9	7.3	6.8	6.5	6.3	6.2	6.1
	14,224	Electricity	–	17.8	17.8	17.8	17.8	17.8	17.8	17.8
Non-Private Car	28,360	Gasoline	12.2	11.0	10.2	9.5	9.1	8.7	8.6	8.6
	33,103	Diesel	9.9	9.0	8.3	7.7	7.4	7.1	7.1	7.0
	26,016	Electricity	–	20.2	20.2	20.2	20.2	20.2	20.2	20.2
Scooter	6,385	Gasoline	4.4	4.3	4.3	4.2	4.1	4.1	4.0	3.9
		Electricity	–	3.7	3.7	3.7	3.7	3.7	3.7	3.7
Heavy Truck	33,388	Diesel	32.0	31.1	30.3	29.4	28.6	27.9	27.1	26.3
Light Truck	11,499	Gasoline	11.5	10.4	9.7	9.0	8.6	8.3	8.2	8.1
	16,719	Diesel	9.8	8.9	8.2	7.6	7.3	7.0	7.0	6.9
Urban Bus	66,084	Diesel	37.6	36.9	36.3	35.6	35.0	34.4	33.8	33.2
		Electricity	–	136.0	136.0	136.0	136.0	136.0	136.0	136.0
Non-Urban Bus	58,793	Diesel	30.7	30.2	29.7	29.1	28.6	28.1	27.7	27.2

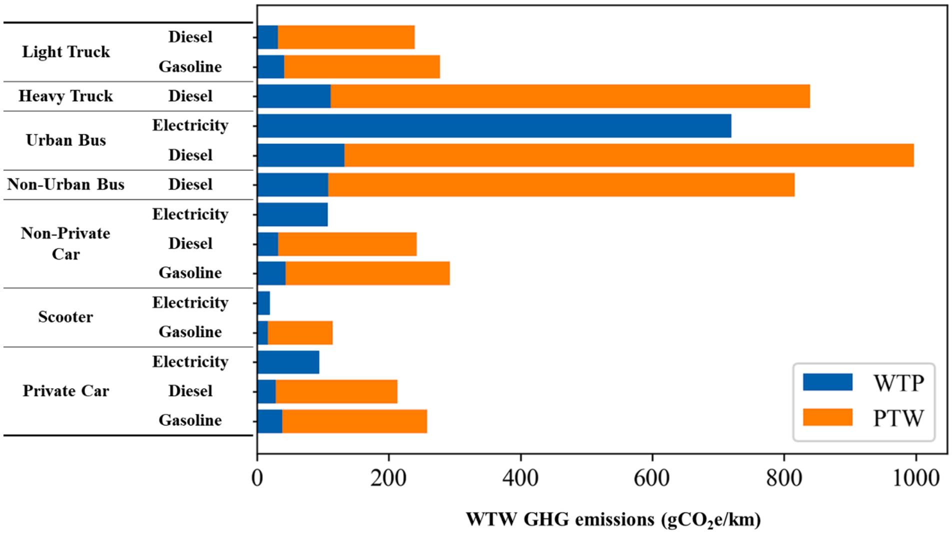


Fig. 6. Fleet-average WTW GHG emissions per kilometer driven by vehicle and fuel type on the road in Taiwan in 2020.

the synergies between clean energy and vehicle electrification, we design another scenario called "EVs with cleaner power grid scenario (i.e., *EV-REN* scenario)." Based on the 2050 Net-Zero power sector roadmap, we project a future trend in the electricity generation mix. First, renewable energy will account for 60 % of Taiwan's power supply in 2050. Second, hydrogen energy, though still not mature nowadays, will be put into operation in 2040 and will increase to 12 % in 2050. Third, pump storage hydropower will continue to provide 1 % of the total gross electricity generation annually. Forth, nuclear power plants, which are still in service today, will be dismantled in 2025. Finally, fossil-fuel power plants will satisfy the remaining electricity demand (i.e., 27 %). For the evolution of Taiwan's fossil-fuel-fired power plants, we account for Taipower's long-term power development plans throughout 2027 (Taiwan Power Company, 2021), including the impacts of new natural gas generator installations and retirements of the existing coal-fired power generators. Fig. 8 summarizes the resulting power mix and the associated electricity GHG emissions intensity applied in the *EV-REN* scenario, reflecting that Taiwan's energy transition highlights a rapid increase in renewables while phasing out nuclear power and substituting coal power with natural gas. Note that the emission benefits of CCUS technology are ignored here due to the unknown emission reduction potentials and the commercialization timelines in Taiwan.

3. Results and discussion

3.1. Vehicle stock projection

Fig. 9 presents the projected median stocks (used as the baseline model predictions later in this study) with 95 % confidence intervals (CIs) obtained using the bootstrap resampling approach for different types of vehicles (see Fig. 10 for the corresponding results of motor vehicles per 100 population). The predicted ranges of possible future private car stock and scooters in Taiwan are 7.38 million

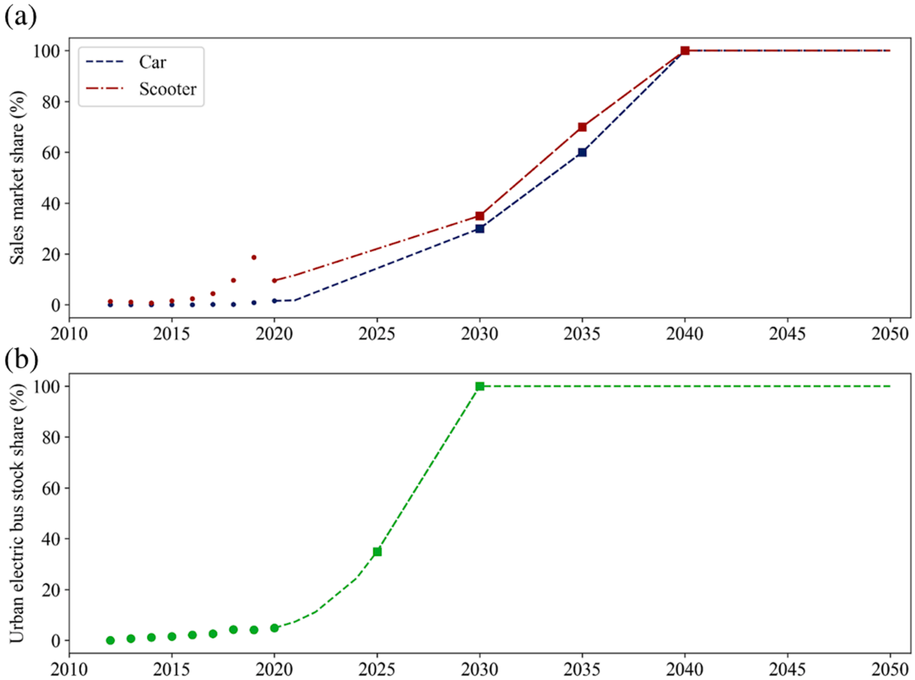
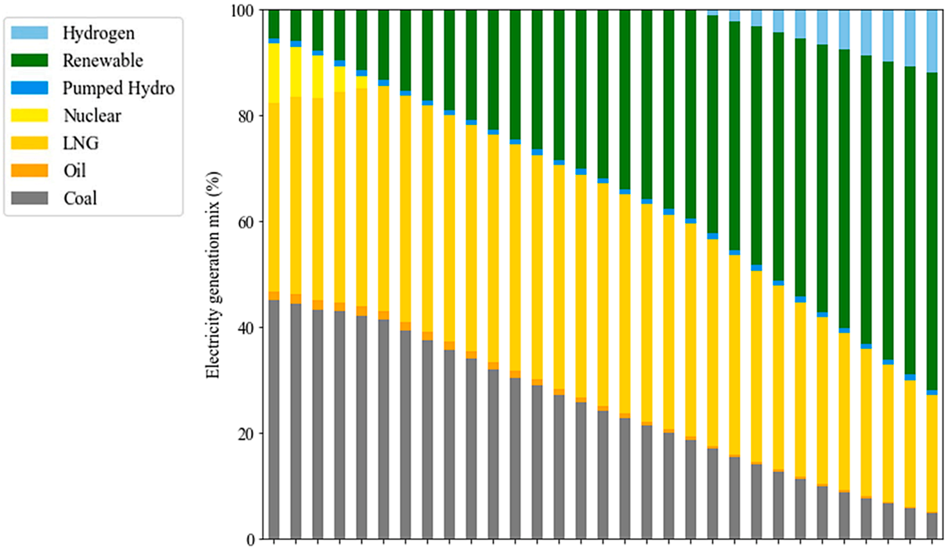


Fig. 7. Historical and projected EV market share from 2015 to 2050 in the *EV* scenario: (a) Annual sales of cars and scooters; (b) Urban bus stock.



Unit	2020	2025	2030	2035	2040	2045	2050
g CO_2e/kWh	530	511	430	357	283	195	116

Fig. 8. Expected power mix and the associated electricity GHG emissions intensity during WTT fuel production stage under *EV-REN* Scenario.

(95 % CI: 7.24 – 7.52 million) and 12.11 million (95 % CI: 11.91 – 12.23 million), respectively, in 2050, together accounting for about 95 % of the total road vehicle stocks. Compared to private cars, scooters, and buses, projecting business car and truck stocks in Taiwan involves greater inherent uncertainty (i.e., wider CIs) owing to the smaller sample size.[3]

[3] As the sample size increases, the variability in the bootstrap distribution decreases.

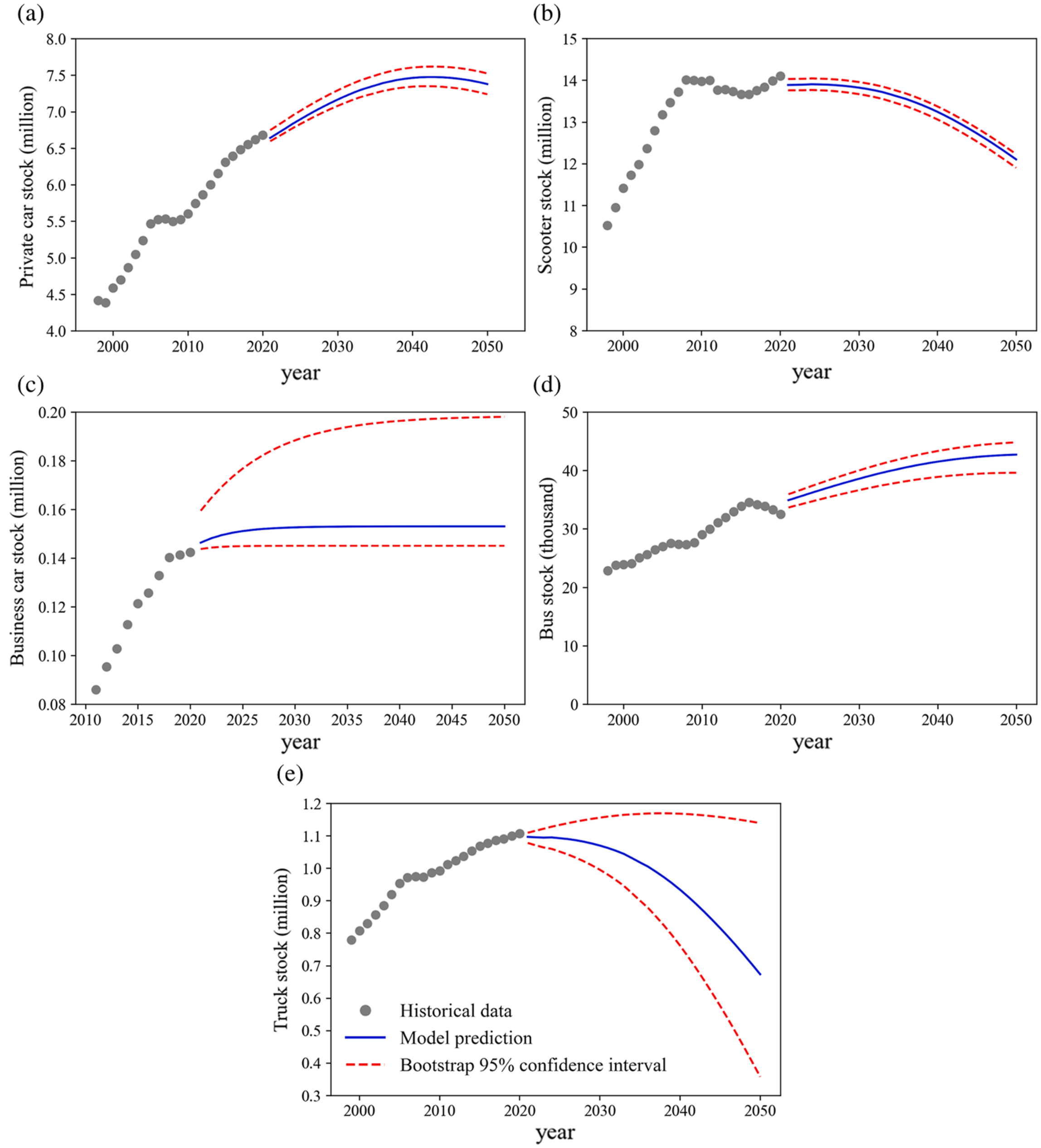


Fig. 9. Vehicle stock projections with bootstrap confidence intervals: (a) Private cars; (b) Scooters; (c) Business cars; (d) Buses; (e) Trucks. Note that the median projections are used as the baseline model predictions.

Fig. 11(a) depicts the projected median results of the vehicle stock in Taiwan over time as EVs diffuse through the market, following the road transport roadmap under the 2050 Net-Zero Pathway. The total number of vehicles on the road will continue to increase gradually until about 22.36 million (95 % CI: 22.02 – 22.77 million) around 2032, then decrease to 21.93 million (95 % CI: 21.45 – 22.48 million) in 2040 and 20.45 million (95 % CI: 19.78 – 21.23 million) in 2050. Despite steady growth in the economy, such an inverted U-shaped pattern of vehicle stock is still expected to occur in Taiwan—primarily due to the shrinking scooter and freight market. The effects of a declining population outweigh the growing GDP per capita on trucking demand, causing the freight market to continue to drop throughout 2050. While scooter and truck contributions to the total vehicle stock will decrease, cars and buses are forecasted to contribute increasingly toward the future (Fig. 11(b)). The current vehicle electrification roadmap will make the nationwide battery-powered car, scooter, and bus stock reach 6.0 million (95 % CI: 5.9 – 6.1 million), 10.3 million (95 % CI: 10.1 – 10.4 million), and 26.3 thousand (95 % CI: 24.4 – 27.6 thousand), respectively, by 2050—corresponding to about 80 % of the total vehicle stock in Taiwan. Fig. 11(c) presents the county-level stock projections in 2050, indicating that most private car and scooter stocks are

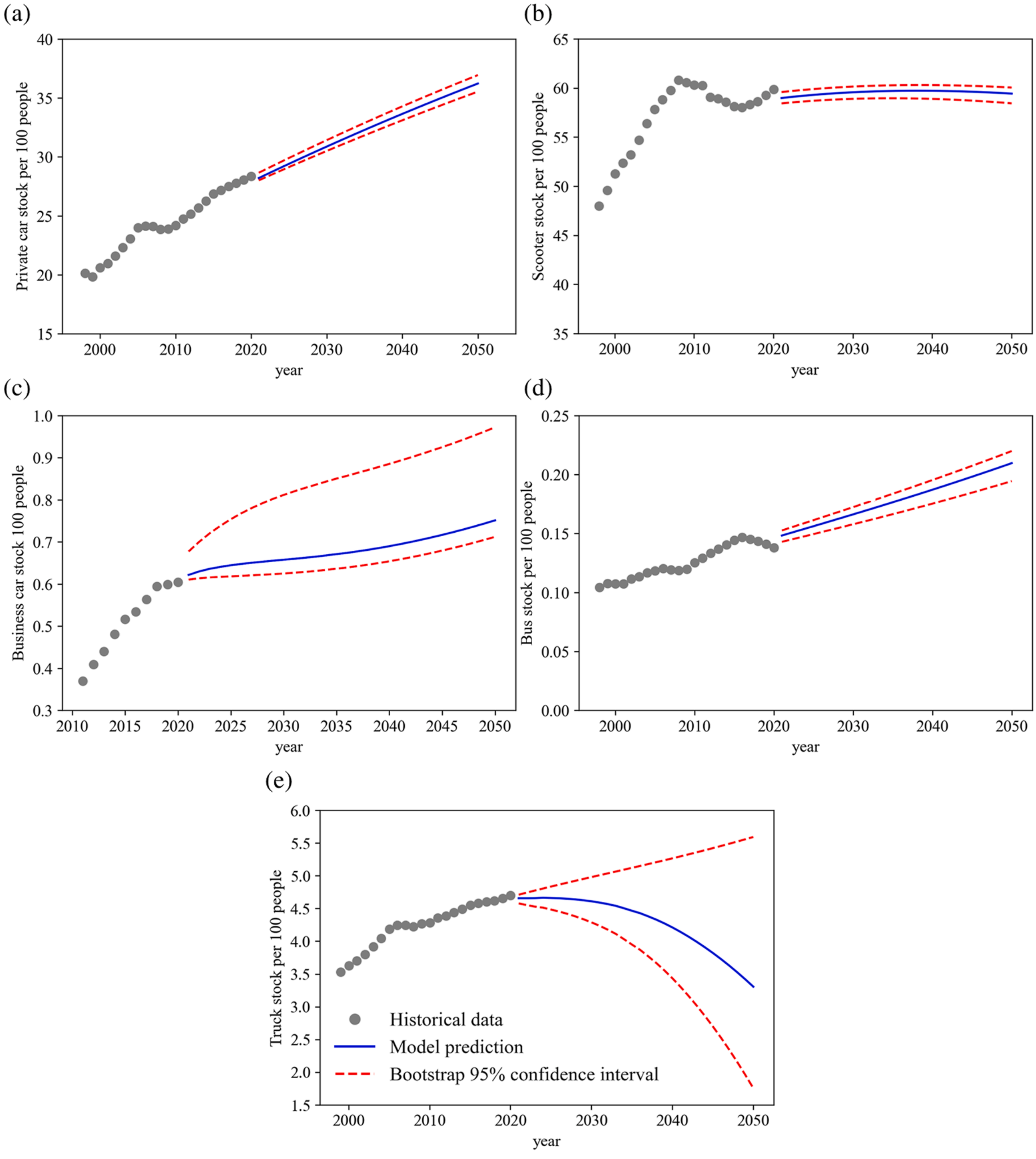


Fig. 10. Vehicle stock per 100 people projections with bootstrap confidence intervals: (a) Private cars; (b) Scooters; (c) Business cars; (d) Buses; (e) Trucks.

in the five most populous metropolitan areas, as expected. However, in terms of ownership rate, these urbanized and densely populated regions generally have lower demand for vehicle ownership (Fig. 11(d)).

3.2. *Life-cycle energy demand and GHG emissions under different scenarios*

Fig. 12 displays the median projections of total energy demand and GHG emissions for the on-road vehicles in Taiwan towards 2050, broken down by vehicle type and powertrain for the *EV* scenario. The median projections with 95 % bootstrap CI are presented in Tables 3 and 4, indicating that the systematic uncertainty will mostly cancel out when estimating the differences between the scenarios. Therefore, we only show the median values for illustrative purposes in the following scenario comparisons. Under all three scenarios, the life-cycle energy demand and GHG emissions will gradually decline from 2020 to 2050 due to the improving ICEVs fuel

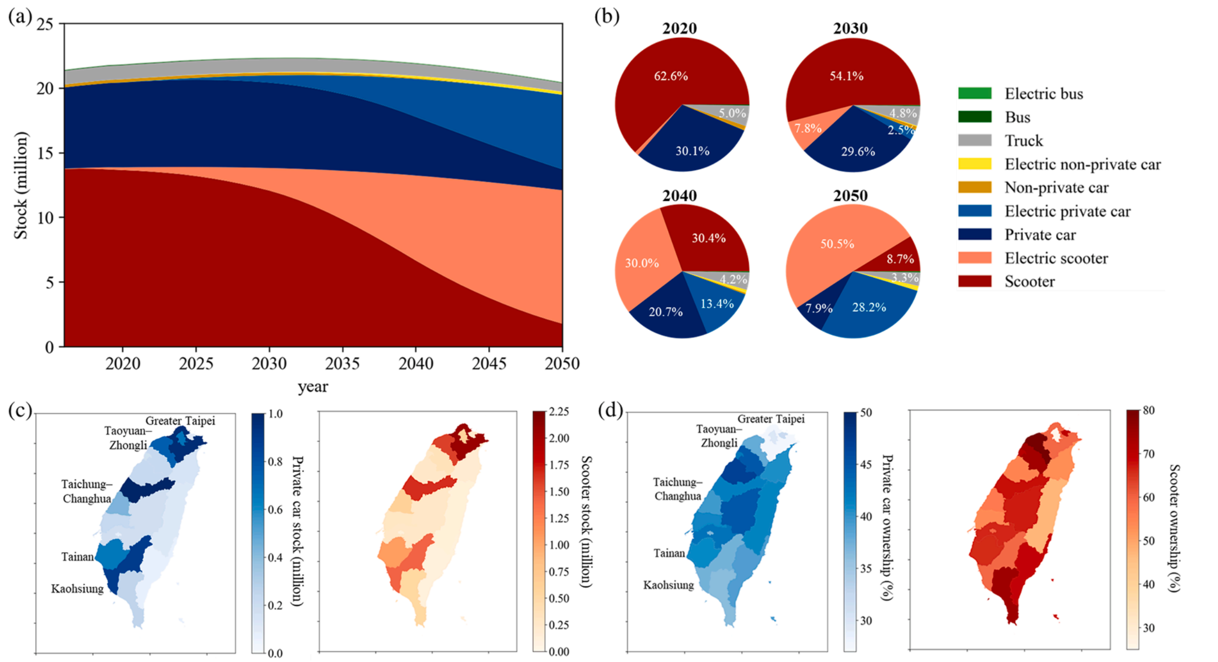


Fig. 11. Projected median vehicle stocks on the road in Taiwan in the *EV* scenario: (a) by type and powertrain from 2020 to 2050; (b) contribution breakdowns among different types and powertrains; (c) and (d) stocks and ownership rates of private car and scooter at the county level in 2050. Note that the *EV-REN* scenario has exactly the same EV and ICEV stock outcomes as the *EV* scenario, while the *REF* scenario has the same total number of vehicles but all powered by ICEs.

efficiency, though the total number of vehicles will keep increasing to the peak in 2031. The overall trend of life-cycle energy consumption is similar to that of the life-cycle GHG emissions. The current vehicle electrification roadmap (i.e., *EV* scenario) will achieve lower GHG emissions compared with the *REF* scenario—the annual life-cycle GHG emissions are reduced by 35.7 % in 2050, and the cumulative reductions from 2020 to 2050 are dropped by 154.4 Mt (13.2 %). Moreover, if combined with cleaner power generation, EV deployment could deliver greater climate change mitigation; the cumulative GHG emissions before 2050 are 78.0 Mt (7.7 %) lower in the *EV-REN* scenario.

As for PTW GHG emissions associated with the direct energy use of on-road vehicles, significantly lower annual GHG emissions are observed under the government planning for road transport electrification. Compared with the *REF* scenario, the *EV* scenario achieves annual direct GHG emission reductions of 11.2 %, 38.1 %, and 69.9 % by 2030, 2040, and 2050, respectively. Despite the substantial GHG emissions reductions, the net-zero target for the road transport sector still cannot be achieved with the current electrification policy framework. According to the 2050 Net-Zero pathway, the road transport sector should be deeply decarbonized and generate only 3.2 Mt carbon emissions by 2050 (Fig. 12(e)). However, the annual direct GHG emissions under the *EV* scenario in 2050 are projected to be 8.4 Mt (95 % CI: 6.8 – 10.7 Mt)—falling far short of what is required to meet the net-zero target.

On the other hand, the increasing market share of EVs shifts the GHG emissions burden from the transport sector to the power generation sector. Since the power grids in Taiwan rely heavily on fossil fuels, the growing electricity generation driven by EV diffusion will inevitably produce more WTP emissions. In the *EV* scenario, with no improvement in grid carbon intensity, the annual WTP GHG emissions of the vehicle fleet in Taiwan will continue to increase through 2050, with the cumulative emissions from 2020 to 2050 increasing by 21.7 Mt (12.7 %) compared with the *REF* scenario. However, with the continuous penetration of renewable energy and substitution of natural gas for coal (i.e., *EV-REN* scenario), the annual WTP GHG emissions will reach a plateau at 7.0 Mt (95 % CI: 6.8 – 7.3 Mt) between 2031 and 2037, then drop to 4.0 Mt (95 % CI: 3.7 – 4.5 Mt) by 2050. Furthermore, the growing WTP GHG emissions from vehicle electrification will gradually be offset by the reductions driven by the cleaner power grid under the power sector decarbonization roadmap to net-zero emissions by 2050.

3.3. Impacts of decarbonization pathways on life-cycle GHG emissions

We break down the cumulative impacts on 2050 life-cycle GHG emissions reductions by various decarbonization pathways proposed in the 2050 Net-Zero Pathway, as shown in Fig. 13. First, electrifying all the operating urban buses by 2030 would cause annual GHG emissions from bus travel to decrease by 9.2 % in 2050, leading to the cumulative carbon reductions of 7.4 Mt. Second, completely phasing out the new sales of ICE cars by 2040 would make the electric car stock in Taiwan reach 6.0 million in 2050, corresponding to nearly 80 % of the total car stock. By 2050, this electric car promotion roadmap would significantly lower annual GHG emissions by 6.5 Mt (34 %), with a cumulative emission drop of 79.6 Mt (12.6 %) compared to the *REF* scenario. Third, banning

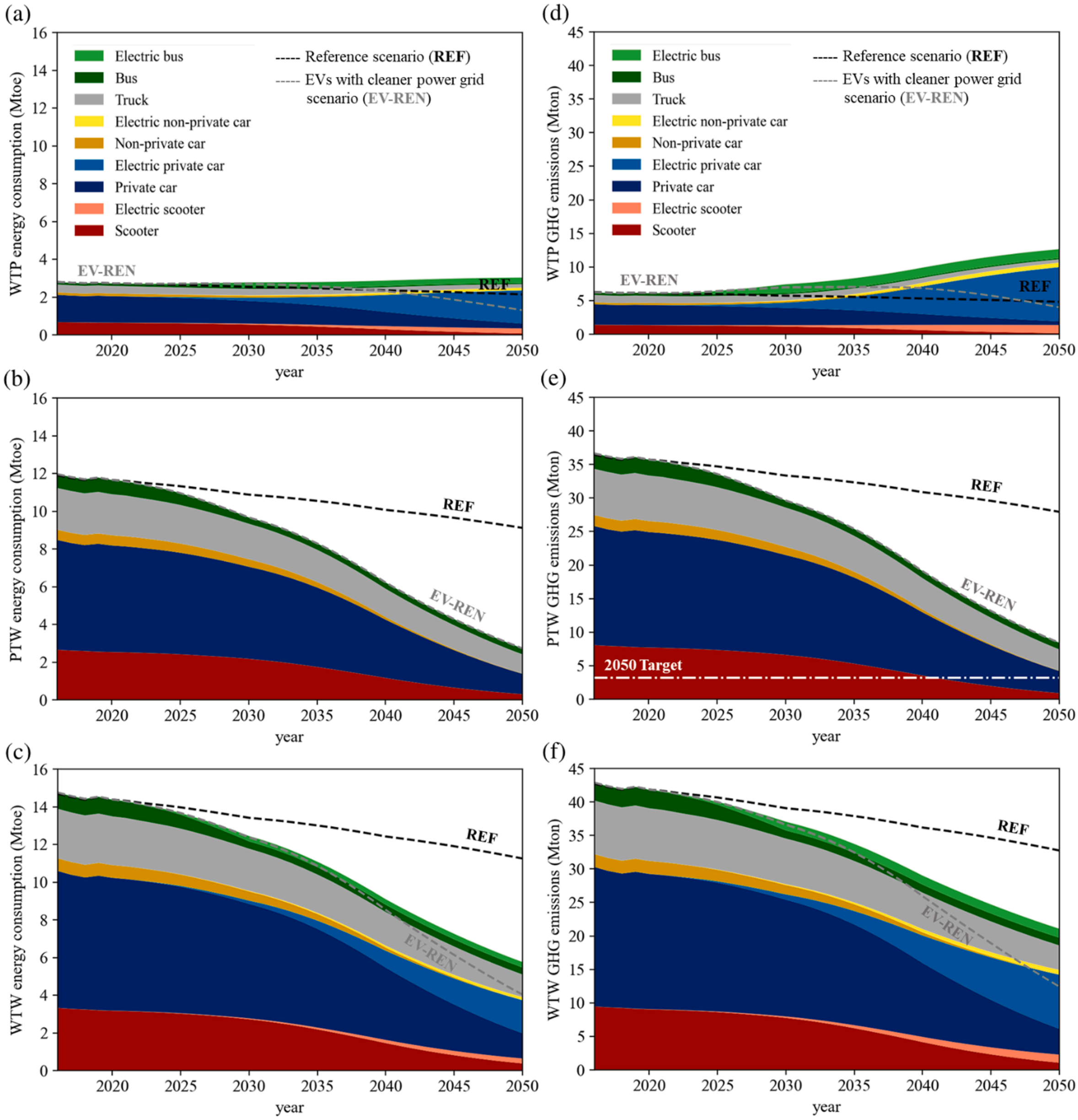


Fig. 12. Scenario analysis results for median WTP, TPW, and WTW (a)-(c) energy demand and (d)-(f) GHG emissions of the on-road vehicles in Taiwan from 2020 to 2050, broken down by vehicle type and powertrain for the *EV* scenario. The horizontal white dashed line in panel (e) indicates the 2050 net-zero target for the road transport sector.

all the new ICE scooter sales by 2040 would lead to 12.1 million electric scooters in 2050, corresponding to 85 % of the total scooter stock. This scooter electrification roadmap would reduce carbon emissions by 4.9 Mt (68.4 %) per year, and the cumulative GHG emissions before 2050 are 67.1 Mt (25.8 %) lower than the *REF* scenario. Finally, decarbonizing the power sector under the 2050 Net-Zero Pathway will have a noticeable impact on life-cycle GHG emissions throughout 2050. An expected reduction in grid emission intensity from 530 g CO_2e/kWh in 2020 to 116 g CO_2e/kWh in 2050 would increasingly lower annual GHG emissions; under the *EV-REN* scenario, the cumulative life-cycle GHG emissions from 2020 to 2050 would decrease by nearly 78.0 Mt (7.7 %) compared with the *EV* scenario.

3.4. Impacts of vehicle use intensity on road transport GHG emissions

Under the current vehicle electrification policy framework, the 2050 net-zero target for the road transport sector still cannot be attained based on the modeling results presented above. The projected road transport GHG emissions rely on the number of vehicles,

Table 3
Energy consumption projections (Mtoe) with 95% bootstrap confidence interval (CI) under three scenarios in Taiwan.

Vehicle Stocks Prediction	Scenario	2020			2030			2040			2050		
		WTW	WTP	PTW	WTW	WTP	PTW	WTW	WTP	PTW	WTW	WTP	PTW
95 % CI Upper	REF	14.60	2.77	11.83	13.86	2.62	11.25	13.16	2.48	10.68	12.38	2.32	10.05
	EV				12.87	2.87	10.00	9.72	3.02	6.71	6.72	3.25	3.47
	EV-REN				12.79	2.80	10.00	9.14	2.43	6.71	4.94	1.47	3.47
Median	REF	14.38	2.73	11.65	13.41	2.54	10.88	12.42	2.35	10.07	11.25	2.14	9.12
	EV				12.44	2.78	9.67	9.09	2.87	6.22	5.76	3.03	2.73
	EV-REN				12.38	2.71	9.67	8.52	2.30	6.22	4.03	1.30	2.73
95 % CI Lower	REF	14.24	2.70	11.54	13.08	2.48	10.60	11.88	2.26	9.62	10.45	2.01	8.44
	EV				12.13	2.71	9.42	8.61	2.76	5.85	5.07	2.87	2.20
	EV-REN				12.07	2.64	9.42	8.06	2.21	5.85	3.39	1.19	2.20

Table 4
GHG emissions projections (Mt CO2e) with 95% bootstrap confidence interval (CI) under three scenarios in Taiwan.

Vehicle Stock Prediction	Scenario	2020			2030			2040			2050		
		WTW	WTP	PTW	WTW	WTP	PTW	WTW	WTP	PTW	WTW	WTP	PTW
95 % CI Upper	REF	42.47	6.22	36.25	40.40	5.92	34.48	38.38	5.62	32.76	36.14	5.29	30.85
	EV				38.32	7.68	30.64	30.99	10.38	20.61	24.13	13.39	10.74
	EV-REN				37.86	7.22	30.64	27.81	7.21	20.61	15.20	4.46	10.74
Median	REF	41.84	6.13	35.71	39.07	5.73	33.34	36.18	5.31	30.87	32.75	4.82	27.94
	EV				37.05	7.43	29.62	29.01	9.91	19.10	21.07	12.66	8.42
	EV-REN				36.60	6.98	29.62	25.93	6.84	19.10	12.42	4.01	8.42
95 % CI Lower	REF	41.43	6.07	35.36	38.07	5.58	32.49	34.54	5.08	29.46	30.33	4.48	25.85
	EV				36.09	7.22	28.86	27.50	9.55	17.95	18.88	12.11	6.78
	EV-REN				35.66	6.79	28.86	24.51	6.56	17.95	10.46	3.68	6.78

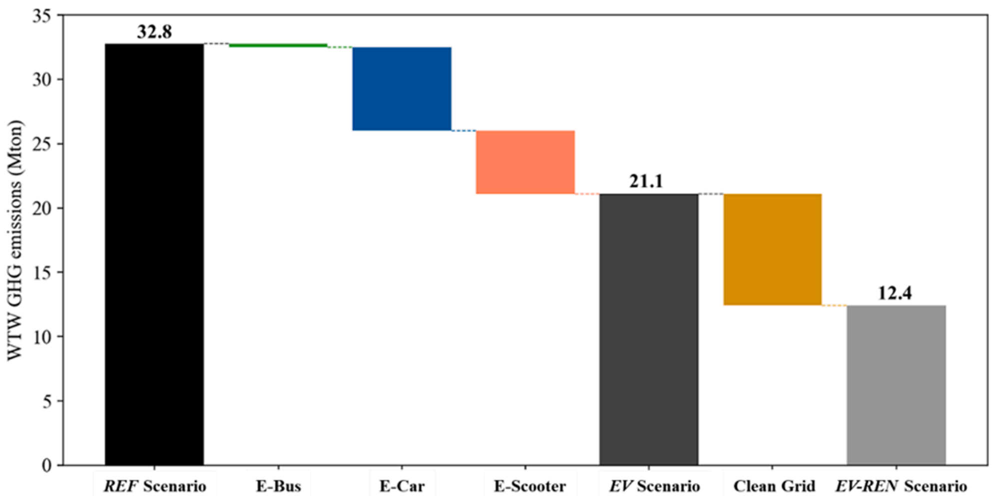


Fig. 13. Reductions in annual life-cycle GHG emissions of the on-road vehicles in 2050 under all three scenarios, broken down by different decarbonization pathways—including electrifying all the operating urban buses by 2030 (E-Bus), all the new car sales by 2040 (E-Car), all the new scooters sales by 2040 (E-Scooter), and switching from coal to renewables (Clean Grid).

fuel consumption rates, and VKT (Eq. (12)); changes in these three factors could significantly affect the model output values. While vehicle stock projections and future fuel consumption rates are mainly based on government forecasts for socioeconomic development and published literature, VKT is assumed to stay constant from 2020 to 2050 due to the lack of relevant data and future trend studies in Taiwan. As the uncertainties in vehicle stock modeling have been investigated by bootstrapping simulation, the influences of VKT assumptions on road transport GHG emissions are studied here.

As a key indicator of vehicle use intensity, VKT is affected by many factors, such as operating costs, trip purpose, public transport network, emerging transportation technologies or services, and management policies. Instead of predicting the future VKT trends based on these factors—which is challenging due to the data limitations, we take the constant vehicle use intensity as the modeling input and conduct the sensitivity analysis to determine the effects of this assumption. We focus the evaluation on private cars and scooters, considering their VKT levels are likely to respond more proactively to the disruption in mobility technologies and services, consumer preferences, and transportation policies.

Fig. 14 presents the GHG emissions change in the road transport sector as the annual VKT levels change. As expected, the PTW GHG emissions increase when VKT increases. However, road transport emissions are more sensitive to vehicle use intensity in the near term than long-term future; this is because a larger fraction of private cars and scooters in 2050 will be electric and thus produce zero tailpipe emissions. The results also suggest that even with the VKT levels reduced by 20 % (e.g., a modal shift from private vehicles to public transport) compared to the current status, the net-zero target for the road transport sector (i.e., 3.2 Mton) still cannot be achieved.

4. Conclusion

The Taiwan Vehicle Fleet Model is developed and applied to study the life-cycle energy consumption and GHG emissions of the on-road vehicle fleet under the ongoing net-zero transition. The model comprehensively characterizes the key features of the road transportation sector in Taiwan, including not only four-wheel vehicles but also two-wheelers fleet turnover pace up to the year 2050. The model also quantifies the parameter uncertainties using the bootstrapping approach and establishes a range of future forecasts in vehicle ownership demands and the corresponding environmental impacts.

The road vehicle stock in Taiwan is projected to peak at 22.4 million (95 % CI: 22.0 – 22.8 million) around 2032 and then decrease to 20.5 million (95 % CI: 19.8 – 21.2 million) in 2050. Despite the steady economic growth, the total number of vehicles would still follow an inverted-U-shaped development trajectory—mainly because of the shrinking population and scooter market. Unlike private car ownership, which will continue to increase to 36.2 % (95 % CI: 35.5 – 37.0 %) through 2050, scooter ownership will reach the peak of the Kuznets curve around 2038 at 59.7 % (95 % CI: 59.0 – 60.3 %). To achieve net-zero emissions by 2050, Taiwan Under the 2050 Net-Zero Pathway (*EV-REN* scenario), vehicle electrification roadmap will lead to widespread adoption of EVs—electric cars, scooters, and buses will together account for nearly 80 % of the total vehicle stock by 2050. the life-cycle GHG emissions of the road vehicle fleets will drop to 12.4 Mt (95 % CI: 10.5 – 15.2 Mt) per year by 2050, accumulating a total of 935 Mt (95 % CI: 898 – 985 Mt) from 2020 to 2050. The overall trends of the direct and life-cycle GHG emissions are similar to energy consumption.

However, the existing vehicle electrification action plans are not sufficient to achieve the GHG emissions reduction targets for the transport sector set by the government. The fleet modeling suggests that the direct GHG emissions of the vehicle fleets under the *EV* scenario will be 8.4 Mt (95 % CI: 6.8 – 10.7 Mt) per year in 2050—significantly lower by 19.5 Mt (70 %) compared with the *REF* scenario, but still 5.2 Mt (163 %) more than the net-zero emissions target for the road transport sector (i.e., 3.2 Mt). The continued EV growth will inevitably shift the GHG emissions from the transport sector to the power sector unless combined with the deep decarbonization of electricity production. The increase in upstream GHG emissions associated with vehicle electrification by 2050 will be offset by the carbon reductions achieved through the clean power sector on the road to a net-zero future.

Based on our analyses, several recommended adjustments to the 2050 Net-Zero Pathway are derived for Taiwan to meet its future climate mitigation targets. First, the timeline to phase out fossil fuel-powered cars and scooters should be more aggressive, and non-urban buses should also be included in the current vehicle electrification roadmap. Suppose all the operating cars, scooters, and non-urban buses are electric in 2050, we find that an additional 5.2 Mt of direct GHG emissions could be avoided, and the decarbonization target for the road transport sector could thus be achieved. Second, sector coupling should be recognized when designing transportation electrification policies. The GHG emissions from the electricity production associated with EV adoption should be assessed in policymaking to accelerate the transition towards a net-zero future. Third, the improving fuel efficiency of new ICEVs should be ensured to decrease carbon emissions in the road transport sector, which is especially important before EVs become cost-competitive.

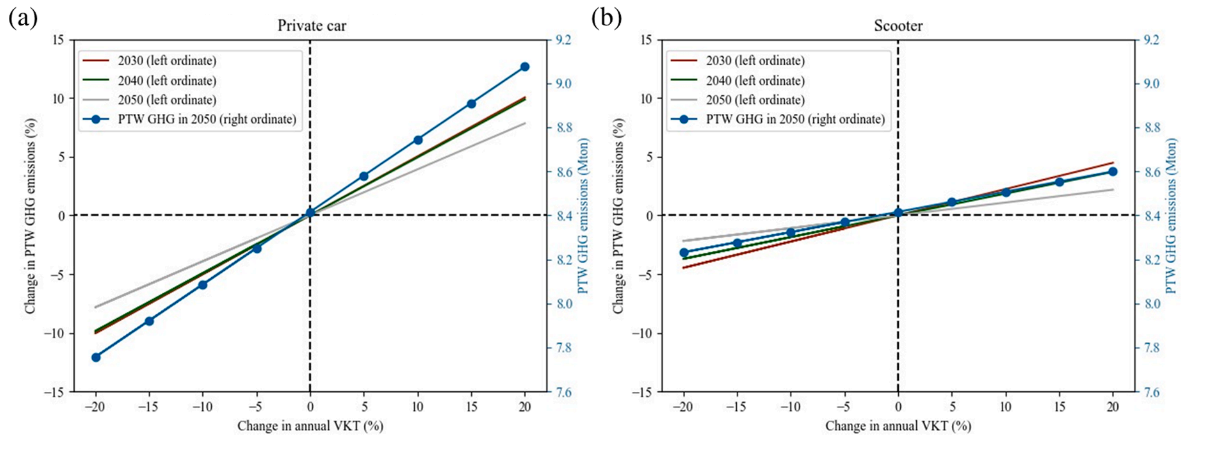


Fig. 14. Sensitivity of the road transport GHG emissions with respect to annual VKT levels (solid lines) and the corresponding PTW GHG emissions in 2050 (lines with circle markers): (a) Private cars; (b) Scooters.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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